

Shen Wang . Jarvis Consulting

Passionate and dedicated software engineer with a solid foundation in computer engineering and diverse hands-on experience across technologies. Demonstrates a strong commitment to continuous improvement in technical expertise, business acumen, leadership, and communication skills. Focused on leveraging technology to solve real-world challenges while evolving as a well-rounded professional capable of bridging technical and non-technical domains.

Skills

Proficient: Python, Langgraph/Langchain, Pyspark, Linux/Bash, RDBMS/PSQL, Agile/Scrum, Git, TypeScript/JavaScript, Tailwind CSS, AWS EC2, RDS, Bedrock

Competent: C#, Java, Next.js, React, GDB, Kafka, Databricks, Pandas, Numpy, Supabase, C/C++

Familiar: Swift, Jenkins, Docker, Wireshark, Protractor, MongoDB, TensorFlow, Unity

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_ShenWang

Real-Time Stock Data Pipeline [GitHub]: Designed and implemented a real-time stock data pipeline using Kafka (Dockerized broker cluster) and PySpark. Built a Python producer (WebSocket connected to a stock data API) and consumer to stream data into Kafka, then performed 5-minute moving average calculations using PySpark for real-time analytics.

Cluster Monitor [GitHub]: Developed an MVP for Jarvis' LCA team to monitor and record server cluster hardware status. Automated data collection of hardware specifications and resource usage using Bash scripts and cron jobs, storing data in a PostgreSQL RDBMS for analysis and resource planning. Utilized Docker for PostgreSQL provisioning and managed version control with Git.

Grep App [GitHub]: Reimplemented **grep** functionality using Java. Created an application that searches through a root directory based on a provided string and saves matches to a user-specified output file. Utilized Docker for distribution, developed with Maven, and optimized with Java Lambda and Stream API.

Highlighted Projects

Smash: iOS Mobile Game: Developed a puzzle-based iOS mobile game in a team of 6. Co-designed game mechanics, coordinated tasks with artists and programmers, and set deadlines to ensure steady development progress. Responsible for the environment generation system and in-game control UI using Unity and C#, leveraging coroutines and event buses.

Professional Experiences

Software Developer, Avazz (July 2024-December 2024): Collaborated within a three-person team to develop an AI-powered data analytics application designed to streamline data consumption by enabling natural language interaction with an AI agent. Independently designed and implemented a data analytics agent workflow using LangGraph and Python, allowing the AI agent to process user questions, generate SQL queries, execute them, and analyze the retrieved data. Integrated asynchronous logic to support concurrent multi-intent question processing. Programmed interactive dashboards to fetch and display insurance data in formats such as pie and line charts, utilizing SQL and TypeScript with server-side rendering (SSR) to enhance security and scalability, and incorporated a Hugging Face transformer model for sentiment analysis. Supported the team by setting up and configuring scalable and robust cloud solutions using AWS RDS and EC2, and leveraging Linux environments to deploy APIs and services. Additionally, explored and tested open-source machine learning models on platforms like Hugging Face, RunPod, and AWS Bedrock, utilizing cloud computing resources to evaluate their potential integration into the project.

Software Developer, Jarvis (May 2024-present): Joined as a trainee and underwent comprehensive project-based training, gaining hands-on experience with various technologies, programming languages, methodologies, databases, and development tools, including MongoDB, PostgreSQL, Maven, Linux CLI, Java, and JavaScript. After a few months, advanced to a role working on a venture project, collaborating with two team members to develop an AI app, contributing to its design and implementation.

Embedded Software Engineer, Beijing Spirit Technology Development Co. (April 2023-January 2024): Integrated a new MCU into existing systems, using AD14 for schematic and PCB layout modifications. Developed and tested peripheral interfaces and assisted in debugging USB connectivity issues.

RPA Developer (Intern), Transport Canada (May 2022-March 2023): Collaborated with clients to gather workflow requirements and propose automation solutions. Developed a UiPath pipeline for file uploads and data recording, significantly reducing manual tasks and saving 133 hours per month.

Full Stack Developer (Intern), Morgan Stanley (January 2021-April 2021): Created a monitoring dashboard for ETL processes and integrated RESTful APIs for data exchange and real-time updates. Improved API query performance with pagination and incremental data loading, reducing page load time by 35%. Contributed to daily scrum meetings and managed monthly deployments using Jenkins.

Education

University of Ottawa (2018-2023), Bachelor of Applied Science - Computer Engineering, Faculty of Engineering - 2020 & 2022 Dean's List honors - Third and fourth-year GPA: 3.7/4

Miscellaneous

- Badminton
- Reading